

Statistics 1: Permutations and Combinations

Syllabus includes:

- Addition and multiplication principles for counting
- Concepts of permutation ($n!$ or nP_r) and combination (nC_r)
- Arrangements of objects in a line or in a circle including cases involving repetition and restriction

1.0.1 Introduction

How many ways can we arrange the letters A, C, and D?

How many ways can you be grouped with 4 other students from your class to form a PW group?

Or, how many ways can a class of 25 students be partitioned into 5 PW groups?

All these questions and more will be addressed in this chapter. Permutations and Combinations are topics in **Combinatorics**, a branch of Mathematics focusing on counting the number of outcomes.

Basic facts good for us to know:

- There are 26 letters in the English alphabet, with 5 vowels (A, E, I, O, U) and 21 consonants.
- There are 52 cards in a standard deck of playing cards, comprising:

4 suits: **Diamonds** ♦ **Clubs** ♣ **Hearts** ♥ **Spades** ♠

Each suit comprises 13 cards: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen and King.

1.0.2 The Addition Principle

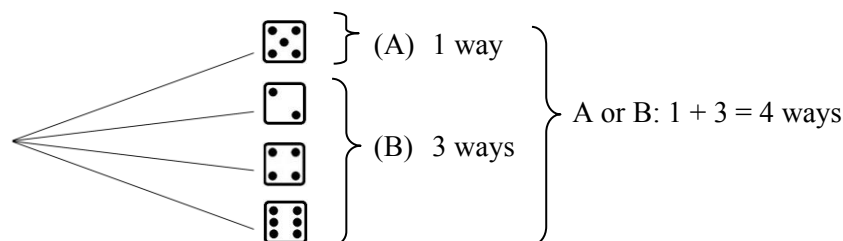
Two operations A and B are **mutually exclusive** if both cannot happen at the same time.

For examples in a die toss, (A) getting a 5 and (B) getting an even number are mutually exclusive.

When two operations A and B are **mutually exclusive**, such that A can happen in r ways and B can happen in s ways, then A or B can happen in $(r + s)$ ways.

In the example used above, there is 1 way of getting a 5 and 3 ways of getting an even number. Hence, the number of ways of getting a 5 or an even number is $1 + 3 = 4$.

Representation in a tree diagram



General result

If there are k mutually exclusive operations and the first operation can be done in n_1 ways, the second in n_2 ways, ..., the k^{th} operation in n_k ways, then one operation or the other can be done in $(n_1 + n_2 + \dots + n_k)$ ways.

Example 1(a)

In order to travel from VJC to Bedok MRT station, there are 4 possible routes by bus and 2 possible routes by taxi. How many routes are there from VJC to Bedok MRT station?

Solution

Since travel routes by bus and taxi are mutually exclusive, the number of routes from VJC to Bedok MRT station is $4 + 2 = 6$.

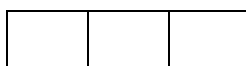
1.0.3 The Multiplication Principle

If a first operation (A) can be performed in r ways, and when it has been performed in any one of these ways, a second operation (B) *can then* be performed in s ways, then the number of ways of performing the two operations (A and B) **successively** is $r \times s$.

E.g. How many ways can we arrange the letters X, Y, Z ?

By listing, there are 6: $XYZ, XZY, YXZ, YZX, ZXY, ZYX$.

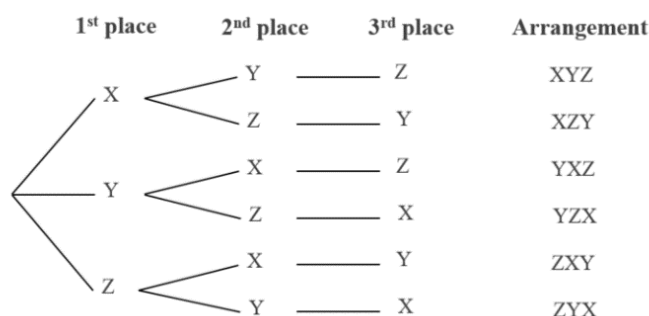
Another way of getting the answer:



We consider 3 places to be filled with the 3 letters, one in each place.

We note that there are 3 ways to fill up the 1st place. After filling up the 1st place (no matter which letter we use), we are left with two other letters to fill up the 2nd place. Then there is only one way of filling up the 3rd place.

Thus the number of arrangements = $3 \times 2 \times 1 = 6$.

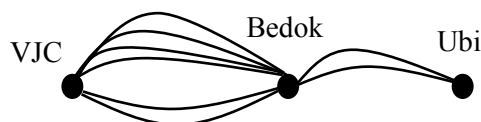
Representation in a tree diagram**Example 1(b)**

There are 6 possible routes from VJC to Bedok MRT station and there are 2 possible MRT routes from Bedok MRT station to Ubi. How many different routes are there from VJC to Ubi via Bedok MRT station?

Solution

The number of routes from VJC to Ubi station is $6 \times 2 = 12$.

[Compare the difference between Examples 1a and 1b.]



General result

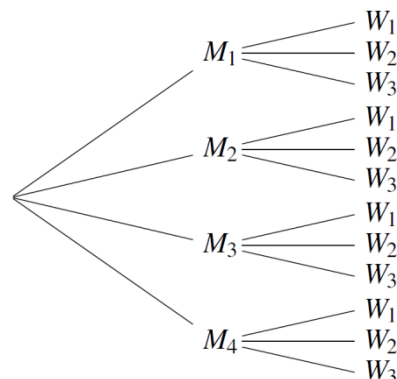
More generally, if k operations are performed in succession and the first one can be performed in n_1 ways, the second in n_2 ways, the third in n_3 ways, ... and the k^{th} operation in n_k ways, then the number of ways of performing the k operations, one following the other, is $n_1 \times n_2 \times n_3 \times \dots \times n_k$.

Example 2

Four men and three women are available to form a mixed doubles pair for a badminton match. How many pairs are possible? Illustrate these outcomes on a tree diagram.

Solution

$$\begin{aligned} \text{Number of ways} &= 4 \times 3 \\ &= 12 \end{aligned}$$

**1.0.4 The factorial notation**

- $n! = n(n-1)(n-2) \times \dots \times 3 \times 2 \times 1$
E.g. $4! = 4 \times 3 \times 2 \times 1 = 24$ – use a calculator (how?)
- $n! = n(n-1)!$
- By definition, $0! = 1$.

Exercise: Without using a calculator, $\frac{10!}{7!} = 10 \times 9 \times 8 = 720$

1.1 Permutations

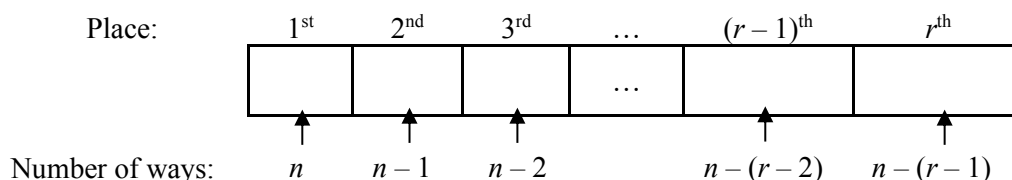
A **permutation** is an ordered arrangement of a number of objects from a given set.

eg. $ACD, ADC, CAD, CDA, DAC, DCA$ are six permutations of the three letters A, C and D .

ADD, DAD, DDA are three permutations of the three letters A, D and D .

1.1.1 Arrangements of r objects taken from n distinct objects

To fill up r places with n distinct objects ($n \geq r$)



The first place can be filled in n ways.

But after filling it, we are left with only $n-1$ objects to fill the 2nd place, and $n-2$ objects to fill the 3rd place.

We then continue with the 4th place, 5th place, ..., until the r^{th} place.

We are left with $n-(r-1)$ objects to fill the r^{th} place.

This is an arrangement with r operations.

By the multiplication principle, the number of ways to fill the r places with n distinct objects = $n(n-1)(n-2)\dots(n-r+1)$.

★ Why do we use the multiplication principle and not the addition principle?

$$\begin{aligned} & \text{Observe } n(n-1)(n-2)\dots(n-r+2)(n-r+1) \\ &= \frac{n(n-1)(n-2)\dots(n-r+2)(n-r+1)(n-r)!}{(n-r)!} \\ &= \frac{n!}{(n-r)!}, \text{ which we denote by } {}^n P_r \end{aligned}$$

As an example,
 $10! = 10 \times 9 \times (8 \times 7 \times \dots \times 1)$
 $= 10 \times 9 \times 8!$

i.e. ${}^n P_r$ = number of permutations of r objects out of n distinct objects.

Note: (a) When $r = n$, ${}^n P_n = \frac{n!}{(n-n)!} = n!$

i.e. the number of permutations of all n distinct objects = $n!$

(b) When $r = 0$, ${}^n P_0 = \frac{n!}{(n-0)!} = 1$

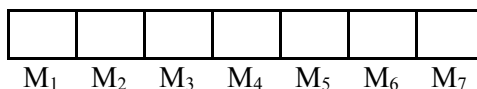
i.e. the number of permutations of 0 objects = 1

Example 3(a) Find the number of permutations of the letters of the word NOISY.

Solution Number of permutations = ${}^5 P_5 = 5! = 120$

Example 3(b) 7 men & 7 women go out on a blind date. In how many ways can they be paired off?

Solution



1st man – can be paired with any of the 7 women, *and then*

2nd man – can be paired with any of the 6 remaining women

⋮

∴ Number of permutations = $7! = 5040$

Common mistake:

$7 \times 7 = 49$ (which gives total number of possible ways to form “one pair”)

★ Contrast with Example 2.

Use a smaller number to understand the mistakes: What about pairing 3 men & 3 women?

Common mistake 1: 3×3 . What does 3×3 mean?

Either M_1, M_2 or M_3 .

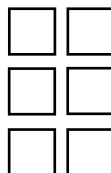


Either W_1, W_2 or W_3 .

This represents the no. of ways to select one pair.

Common mistake 2: $3! \times 3!$. What does $3! \times 3!$ mean?

Permute the 3 men on the left.



Permute the 3 women on the right.

These two set of pairings are the same.

(M_1, W_1) (M_2, W_2)
 (M_2, W_2) and (M_3, W_3)
 (M_3, W_3) (M_1, W_1)

Example 4(a) Find the number of 5-letter codes which can be formed from the letters of FRIENDS.

Solution

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In first place, we have 7 choices,
And then in second place, we have 6 choices.

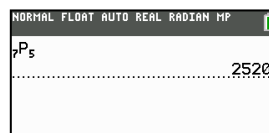
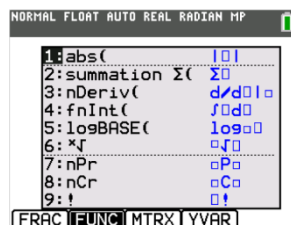
⋮

Number of codes = $7 \times 6 \times 5 \times 4 \times 3$

$$= {}^7P_5 = 2520$$

(or just see it as arranging 5 out of 7 distinct objects)

Use of GC to find nP_r



You can access the nP_r function, using **alpha** **window** (option 7).

Example 4(b) In how many ways can 5 students be given one book each if there are 8 different titles available?

Solution

1st student: any of the 8 books

And then 2nd student: any of the 7 remaining books, etc.

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Number of ways = ${}^8P_5 = 6720$

Example 4(c) In how many ways can 6 policemen be assigned to 12 coaches of a train if each officer must ride in a different coach?

Solution

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No. of ways = ${}^{12}P_6 = 665280$

Remember: Do not just understand the given solutions, but also understand why your other methods are valid/invalid.

1.1.2 Permutations with restrictions

Generally, we solve such questions by *first* dealing with the restrictions.

Example 5 Three men and three women are to sit in six seats arranged in a line. Two of the women are named April and June. In how many ways can this be done if

- there are no restrictions,
- no two men and no two women may sit together,
- April and June must sit next to each other,
- April and June must not be adjacent?

Solution

(a) Number of ways = $6!$
 $= 720$

(c) $\boxed{A J}$ _ _ _ _

(b) MWMWMW or WMWMWM

Process

1. Arrange the 3 Men ($3!$ ways)
2. *And then* arrange the 3 Women ($3!$ ways)
3. Apply Addition Principle to the two mutually exclusive cases

Number of ways = $3!3! + 3!3!$
 $= 72$

Process

1. Group April and June as 1 unit
2. Arrange 5 units ($5!$ ways)
3. *And then* arrange April and June within big unit ($2!$ ways)

No. of ways = $5!2!$
 $= 240$

(d) No. of ways (= total no. of ways – no. of ways when they are together)

= $720 - 240$
 $= 480$

In part (b), do you know when to $\times$ and when to $+$?

Complementary events

‘April & June are adjacent’ and ‘April & June are not adjacent’ in part (d) are known as **complementary events**.

Two events are **complementary** if they are

- (i) they are mutually exclusive, and
- (ii) their union covers all possible outcomes.

Example 6 In how many ways can the letters of the word ALTRUISM be arranged in a row if
 (a) the three vowels are together,
 (b) the first and last letters are consonants,
 (c) all three vowels are separated?

Solution

(a) AUI _ _ _ _

1. Group 3 vowels as one unit
2. Arrange 6 units
3. *And then* arrange 3 vowels within unit

$$\begin{aligned}\text{No. of ways} &= 6!3! \\ &= 4320\end{aligned}$$

(b) _ _ _ _ _

1. Arrange 2 (out of 5) consonants in first and last places
2. *And then* arrange remaining 6 letters

$$\begin{aligned}\text{No. of ways} &= {}^5P_2 \times 6! \\ &= 14400\end{aligned}$$

(c)

$\begin{array}{cccccc} \text{L} & \text{T} & \text{R} & \text{S} & \text{M} & \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \end{array}$

1. Arrange 5 consonants first
2. *And then* insert the 3 vowels among any 3 of the 6 slots to separate them.

$$\begin{aligned}\text{No. of ways} &= 5! {}^6P_3 \\ &= 14400\end{aligned}$$

Note: Settle restrictions first.

First blank (5 ways), then last blank (4 ways), followed by the remaining 6 letters (6! ways). i.e. $5 \times 6! \times 4$.

THINK: Is “3 vowels not together” equivalent to “all 3 vowels are separated”?
 i.e. Is no. of ways = total no. – (a)?

“Insertion” Method

- 1) Identify items to separate and the separators
- 2) Arrange the separators first, then insert the items that need to be separated.

Example 7 Find the number of ways of arranging 6 men and 2 boys in a row if there are
 (a) exactly 2 men separating the boys,
 (b) at least 2 men separating the boys.

Solution

(a) — BMMB — — —

Arrange 2 men out of 6 men between boys
 and then arrange 2 boys.

These 2 men and 2 boys form 1 unit,
 and then arrange 5 units.

$$\begin{aligned} \text{No. of ways} &= {}^6P_2 \times 2! \times 5! \\ &= 7200 \end{aligned}$$

(b) Method 1:

Consider 2 men separating boys, 3 men separating boys, 4 men separating boys, 5 men separating boys, 6 men separating boys.

Too many cases! Bad method.

Method 2: (Complement Method)

Consider taking the complement of:
 1 man separating boys or 0 man separating boys.

ie. BMB M M M M M
BB M M M M M

$$\begin{aligned} \text{No. of ways} &= 8! - 6 \times 2! \times 6! - 7!2! \\ &= 21600 \end{aligned}$$

Example 8 How many numbers less than 5000 can be formed using some or all of the integers 1, 4, 5, 7, with no digit being repeated?

Solution

No. of digits in the number	1	2	3	4
No. of ways	— 4	— — 4 × 3	— — — 4 × 3 × 2	— — — — 2 × 3 × 2 × 1

$$\therefore \text{Total number of ways} = 4 + 12 + 24 + 12 = 52$$

Why do we consider the cases separately, then add up the number of ways?

1.1.3 Permutations with repetitions of objects

Example 9 How many 4-digit numbers can be formed using the digits 1, 3, 4, 5, 8 if
 (i) no digit can occur more than once,
 (ii) repetition of digits is allowed.

Solution

(i) Number of ways = ${}^5P_4 = 120$

(ii) Number of ways = $5 \times 5 \times 5 \times 5$
 $= 5^4 = 625$

Common errors: 5×4 and 4^5 .

One way to avoid such errors is to draw out the blanks “ ” for the 4-digit number, and consider how many possibilities for each blank.

- Example 10** Find the number of ways 4-digit numbers greater than 3 000 can be formed using the digits 0, 1, 3, 4, 5 if
- no repetition of digits is allowed,
 - no repetition of digits is allowed and the number is an even number,
 - repetition of digits is allowed.

Solution

(i) — — — —

1st place: 3, 4 or 5, *and then*

next 3 places: remaining 4 digits.

Number of ways = $3 \times {}^4P_3 = 72$

(ii) — — — — **or** — — — —

Case 1 – 1st place: 3 or 5, last place: 0 or 4,
remaining 2 places: remaining 3 digits.

Case 2 – 1st place: 4, last place: 0,
remaining 2 places: remaining 3 digits.

Number of ways

= $(2 \times 2 \times 3 \times 2) + (1 \times 1 \times 3 \times 2) = 30$

(iii) — — — —

Number of ways = $3 \times 5 \times 5 \times 5 - 1 = 374$

Why is considering cases here?

What if we considered the following?

Case 1 – last place is 0;

Case 2 – last place is 4.

Again, check that you know when to use $\times$
and when to use $+$.

- Example 11** A multiple-choice test consists of 10 questions, each with four options, of which one is the correct answer. In how many ways can Alan, who is totally unprepared for the test, answer all the questions? Assume he chooses one option for each question.

Solution

Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	Q ₆	Q ₇	Q ₈	Q ₉	Q ₁₀

Number of ways = $4^{10} = 1048576$

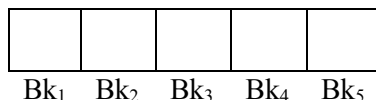
Common errors: 10×4 and 10^4 .

Again, you can avoid such errors by drawing out the blanks or boxes shown “— — — —” for the 10 questions, and think of the number of possibilities for each blank.

Compare this with Example 9 (ii).

Example 12 Find the number of ways in which 5 different books can be distributed among 8 people if each person can get any number of books. All 5 books must be distributed.

Solution



1st book : to any of the 8 persons
 2nd book : also to any of the 8 persons
 (not 7 remaining persons)
 ⋮
 5th book : to any of the 8 persons

Number of ways = $8^5 = 32768$

Common errors: 5×8 and 5^8 .

Contrast with Example 4 (b).

Depending on the question, are you able to figure out what should the boxes/blanks represent? (The books or the people?)

1.1.4 Arrangements of n objects, not all distinct

Let's suppose we want to arrange the letters of the word *MOO*.

Let's label the two *O*'s with suffixes, and then arrange *M*, *O*₁, *O*₂ as three distinct objects.

Total number of arrangements = $3!$:

<i>M O</i> ₁ <i>O</i> ₂	<i>M O</i> ₂ <i>O</i> ₁
<i>O</i> ₁ <i>M O</i> ₂	<i>O</i> ₂ <i>M O</i> ₁
<i>O</i> ₁ <i>O</i> ₂ <i>M</i>	<i>O</i> ₂ <i>O</i> ₁ <i>M</i>

But actually, *O*₁ *O*₂ and *O*₂ *O*₁ are not distinguishable, and form only one arrangement: *OO*.

Thus the total number of arrangements = $\frac{3!}{2!}$

In general, number of permutations of n objects, of which r are alike = $\frac{n!}{r!}$.

Similarly, number of permutations of n objects, of which n_1 are alike of the 1st kind, n_2 are alike of the 2nd kind, ..., n_r are alike of the r th kind is

$$\frac{n!}{n_1! n_2! \dots n_r!}.$$

Example 13 All the letters of the word EVERMORE are arranged. Calculate the number of different ways such that

(i) there is no restriction, (ii) the first letter is E, (iii) all 3 E's are consecutive?

Solution EEE V RR M O

(i) 8 letters, of which 3 are identical E's, 2 are identical R's

Number of ways = $\frac{8!}{3!2!} = 3360$

(ii) E _ _ _ _ _

1) 1 way (not 3) to put E in the first place
 2) Arrange the remaining 7 letters
 EEVRRMO of which 2 are identical E's & 2 are identical R's.

Number of ways = $1 \times \frac{7!}{2!2!} = 1260$

(iii) _ [EEE] _ _ _ _

Group EEE as 1 unit and then arrange 6 units of which there 2 are identical R's.

Number of ways = $\frac{6!}{2!} = 360$

Check Point A: Try this

1. Find the number of different arrangements of the six letters in the word ELEVEN in which
 - (i) there are no restrictions;
 - (ii) all three letters E are consecutive;
 - (iii) the first letter is E and the last letter is N;
 - (iv) the letters V and N are separated;
 - (v) all three letters E are separated.

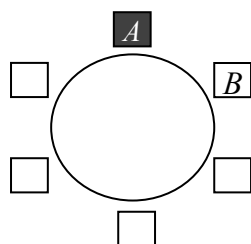
1.1.5 Arrangements in a circle

When we arrange 6 objects A, B, C, D, E, F in a row, A can be arranged in 6 ways (ie. any one of the 6 positions), and then B in 5 remaining ways, ... thus the number of arrangements = $6 \times 5 \times 4 \times 3 \times 2 \times 1$.

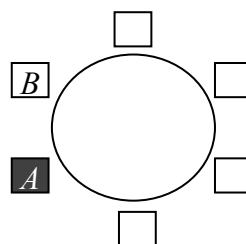


However, when we arrange them in a circle, A can be arranged in only 1 way, because all the 6 positions around a circle are not distinguishable (there is no “first place” nor “second place” etc). And then once A has taken up a position, there are now 5 ways to arrange B (one place to the left of A , or two places to the left, or, ...). Similarly, C has 4 ways, etc.

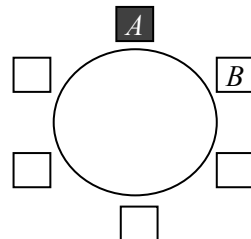
$\therefore$ the number of arrangements = $1 \times 5 \times 4 \times 3 \times 2 \times 1$.



is not different from

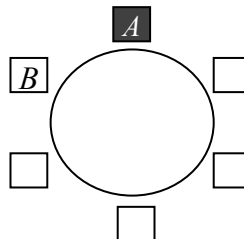


, i.e. still 1 way.



But

is different from

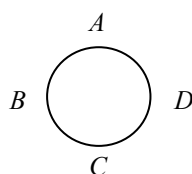
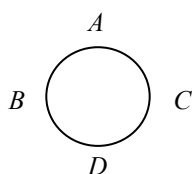
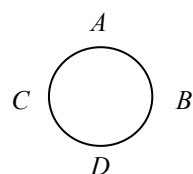
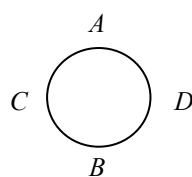
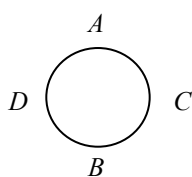
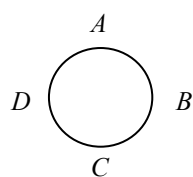


, i.e. 2 ways here.

$\therefore$ In general, number of ways to arrange n distinct objects in a circle = $1 \times (n-1)!$

There is only 1 way to arrange the first object, and then the remaining $(n-1)$ objects can be arranged in $(n-1)!$ ways.

Example: The following are the $1 \times 3! = 6$ circular arrangements of A, B, C and D .



- Example 14** Find the number of ways in which 4 boys and 4 girls can be seated round a table if
- there is no restriction,
 - no two people of the same gender are to sit next to each other,
 - one particular girl must sit between two particular boys,
 - one particular girl must sit between any two boys.

Solution

(i) Number of ways = $1 \times 7! = 5040$

(ii) 1) Arrange 4 boys in a circle in $3!$ ways.

2) And then arrange 4 girls in $4!$ ways.

$\therefore$ Number of ways = $1 \times 3! \times 4! = 144$

(iii) 1) Group the two particular boys and the particular girl as a unit

2) Arrange the 6 units in a circle

3) and then arrange the particular boys in $2!$ ways

Number of ways = $1 \times 5! \times 2! = 240$

(iv) 1) Fill up the two places next to the particular girl with 4 boys in 4P_2 ways

2) Group the two boys and the particular girl as a unit

3) And then arrange the 6 units (big unit and 5 remaining people) in a circle

Number of ways = ${}^4P_2 \times 1 \times 5! = 1440$

THINK: For part (iv), are you able to do it in the following manner?

1) Sit the boys down in $1 \times 3!$ ways, then

2) sit the particular girl down in between any two boys in 4 ways, then

3) sit the other 3 girls down in $3 \times 4 \times 5$ ways.

It would be useful for you to explore various methods to check and understand why some methods are feasible while others are not. Compare different methods with your friends and try to explain the thought process when completing the tutorial questions.

1.2 Combinations

1.2.1 Combinations of r objects taken from n distinct objects

A **combination** is a selection of a number of objects from a given set, disregarding the order of selection.

Number of combinations of r objects taken from n distinct objects ($n \geq r$) is denoted by ${}^n C_r$ or $\binom{n}{r}$.

Illustration:

Consider the combinations of three objects taken from five objects denoted by A, B, C, D and E .

<u>Combination</u>	<u>Permutations</u>
ABD	$ABD, ADB, BAD, BDA, DAB, DBA$
BCE	$BCE, BEC, CBE, CEB, EBC, ECB$
$\vdots$	$\vdots$

Observe: each combination gives $3!$ permutations.

$\therefore$ Total number of combinations $\times 3! =$ total number of permutations ($= {}^5 P_3$)

ie. Total number of combinations $= \frac{{}^5 P_3}{3!}$

In general, we know that number of permutations of r objects taken from n distinct objects $= {}^n P_r$.

But we can also view permutations of r objects taken from n objects in 2 steps:

We **choose** the r objects first without arranging them – there are ${}^n C_r$ ways.

And then for each combination of r objects, we arrange all r objects – in $r!$ ways.

So total number of ways $= {}^n C_r \times r!$

Thus

$${}^n C_r \times r! = {}^n P_r$$

Number of combinations of r objects taken from n distinct objects, is

$${}^n C_r \left[\text{also } \binom{n}{r} \right] = \frac{{}^n P_r}{r!} = \frac{\frac{n!}{(n-r)!}}{r!} \Rightarrow {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$\text{E.g. } {}^{19} C_4 = \frac{19!}{4!15!} = \frac{19 \times 18 \times 17 \times 16}{1 \times 2 \times 3 \times 4}$$

A useful result

The number of combinations of r objects from n distinct objects is the same as the number of combinations of $(n-r)$ objects from n distinct objects – since each time we select r objects, $(n-r)$ objects are left behind.

$$\therefore {}^n C_r = {}^n C_{n-r}$$

Another proof (algebraic):

$$\text{e.g. } {}^9 C_3 = {}^9 C_6$$

$${}^n C_{n-r} = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!} = {}^n C_r$$

$$\text{e.g. } {}^n C_0 = {}^n C_n = 1$$

$$\text{e.g. } {}^n C_1 = {}^n C_{n-1} = n$$

Recap:

A **permutation** is an **ordered** arrangement of a number of objects from a given set.

A **combination** is a selection of a number of objects from a given set, disregarding the order of selection.

- Example 15** Given a class of 3 girls and 4 boys, how many committees of 5 can be formed if
- there are no restrictions,
 - if the committee is made up of 2 girls and 3 boys,
 - if the committee must have at least 2 girls,
 - if the committee must have at least 2 girls and at least 2 boys?

Solution

(i) Number of ways = ${}^7C_5 = 21$

(ii) Number of ways = ${}^3C_2 \times {}^4C_3 = 12$

(iii) Method 1: (Direct Method)

2G3B (from (ii)) or 3G2B

$\therefore$ Number of ways

$$= 12 + {}^3C_3 \times {}^4C_2$$

$$= 18$$

Method 2: (Complement Method)

Complement: 0 girl or 1 girl

(0 girl not possible. Why?)

$\therefore$ number of ways = $(i) - {}^3C_1 \times {}^4C_4$
18

Common mistake for (iv): Choose 2 girls & 2 boys first, and then 1 more of any gender from remaining 3, getting the wrong answer of ${}^3C_2 \times {}^4C_2 \times {}^3C_1$.

What's wrong?

(iv) 2 cases: Either 2G3B or 3G2B

$\therefore$ number of ways = (iii) = 18

- Example 16** A committee of 3 people is to be formed from 5 couples. Find the number of ways this committee can be formed if
- the committee must consist of at least 1 woman,
 - any husband and wife cannot both serve in the committee.

Solution

- (i) Number of ways

$$= {}^{10}C_3 - {}^5C_0 \times {}^5C_3$$

$$= 110$$

(i): *Complement*: What is the complement?

Direct: What are the cases that you should consider for a direct method?

- (ii) Method 1: Direct Method

Number of ways

$$= {}^5C_3 + {}^5C_2 \times {}^3C_1 + {}^5C_2 \times {}^3C_1 + {}^5C_3$$

$$= 80$$

(ii): *Direct*: 4 Mutually exclusive cases (3 men) or (3 women) or (select 2 men first and select 1 woman out of 3 women who are not wives of these 2 men) or (select 2 women first and select 1 man out of 3 men who are not husbands of these 2 women).

Method 2: Complement Method

Number of ways

$$= {}^{10}C_3 - {}^5C_1 \times {}^8C_1$$

$$= 80$$

Complement: "Selecting 1 couple out of 5 couples, then selecting 1 person out of the 8 remaining people."

- Example 17** In how many ways can 4 letters be selected from the letters in the word SLEEPY?

Solution

Case	No. of ways
A B C D [for case 1 and 2]	5C_4
A B E E [for case 3]	${}^4C_2 \times 1$
A E E E [for case 4]	${}^4C_1 \times 1$

Answer is not simply 7C_4 . Why?

How else would you be able to split into cases? (Number of E's selected?)

Case 1: 0 E

Case 2: 1 E

Case 3: 2 E's

Case 4: 3 E's

Total number of ways = $5 + 6 + 4 = 15$

Check Point B: Try this

- Find the number of 4-letter code-word that can be made from the letters of the word ADVANCE,
 - using none of the "A"s,
 - using both of the "A"s.
 How many 4-letter code-words can be form using any 4 letters from the word ADVANCE.

1.2.2 Partitioning n objects**Example 18** In how many ways can 6 men be divided into 3 groups, consisting of

- (i) 3 men, 2 men and 1 man respectively,
- (ii) a first pair, a second pair and a third pair,
- (iii) 2 men in each group,
- (iv) 4 men, 1 man and 1 man respectively?

Solution

(i) Number of ways

$$= {}^6C_3 \times {}^3C_2 \times {}^1C_1$$

$$= 60$$

(i): choose 3 men for the first group, *then* 2 men out of the remaining 3 people for the second group etc.

THINK: Will ${}^6C_1 \times {}^5C_3 \times {}^2C_2$ give the same answer?
i.e. Does it matter which group you form first?

(ii) Number of ways

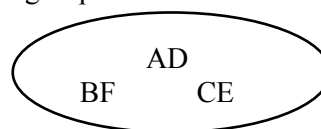
$$= {}^6C_2 \times {}^4C_2 \times {}^2C_2$$

$$= 90$$

Note that in part (ii), these count as two different ways:

1st Pair	2nd Pair	3rd Pair
AD	BF	CE
AD	CE	BF

However, these don't form 2 ways in part (iii), as we still have the same 3 pairs of people. In part (iii), the pairs of people form a group without order.



We see that each way in (iii) will give rise to 3! ways in (ii).

$\therefore$ Total number of ways in (iii) $\times$ 3! =
Total number of ways in (ii)

(iii) Number of ways

$$= \frac{90}{3!}$$

$$= 15$$

(iv) Number of ways

$$= \frac{{}^6C_4 \times {}^2C_1 \times {}^1C_1}{2!}$$

$$= 15$$

(iv): Divide by 2! because the last two groups (individual men) are not distinguishable.

The expression ${}^nC_a \times {}^{n-a}C_a$ refers to the number of selections of a items for the first group, followed by selecting a items from the remaining items for the second group.

However, when the two groups are not distinguishable, then the number of ways is $\frac{{}^nC_a \times {}^{n-a}C_a}{2!}$.

You can generalise both results above for three or more groups.

For example, consider the case when we divide cards A, B, C and D into 2 pairs.

Using ${}^4C_2 \times {}^2C_2 = 6$, the 6 outcomes are

1. AB CD;
2. AC BD;
3. AD BC;
4. BC AD;
5. BD AC;
6. CD AB.

However, there are repeated pairs such as ((1) & (6)), ((2) & (5)), and ((3) & (4)). The correct answer is given by $\frac{{}^4C_2 \times {}^2C_2}{2!} = 3$.

The 3 ways are listed as: (AB CD), (AC BD), and (AD BC).

1.2.3 Combinations of n objects (distinguishable) taken any number at a time

Example 19 How many selections can be made from 4 different books, if we select at least one book?

Solution

Method 1: Number of selections = ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 15$

Method 2: 1st book: selected or not selected (ie. 2 ways)

2nd book: selected or not selected (ie. 2 ways)

3rd book: selected or not selected

4th book: selected or not selected

Number of selections = $2 \times 2 \times 2 \times 2 - 1 = 15$

Example 20 How many possible subsets can be formed from a set of 6 elements?

Solution

Method 1: Number of possible subsets = ${}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6 = 64$

Method 2: Number of possible subsets = $2^6 = 64$

(i.e. For each element, select it or do not select it for the subset, ie. 2 cases)

1.2.4 Combinations of n objects (not all distinguishable) taken any number at a time

Example 21 How many different sums of money can be formed by using some or all of the following: 3 one-dollar coins, 1 five-dollar note and 5 ten-dollar notes?

Solution

One-dollar coin – take none or 1 or 2 or 3: 4 ways

Five-dollar note – take none or 1: 2 ways

Ten-dollar note – take none or 1 or 2 or ... or 5: 6 ways

Number of ways = $4 \times 2 \times 6 - 1 = 47$

Note: Method does *not* work if there are 3 one-dollar coins, 4 five-dollar notes and 5 ten-dollar notes. Why?

1.2.5 Combinations with permutations

Example 22 Four girls and a boy, selected from 7 girls and 3 boys, are to be seated in a row. Find the number of ways this can be done if the boy must always be in the centre.

Solution

Select the 4 G and 1 B first from 7 G and 3 B. (No. of ways = ${}^7C_4 \times {}^3C_1$)

Then arrange the 4 G with B in the centre. $\therefore$ No. of ways = ${}^7C_4 \times {}^3C_1 \times 4!$
= 2520

Examples 4(a), (b), (c) can also be solved using a similar method (select first, then arrange; rather than permute the items directly).

Example 23 A box contains 7 beads: 3 red, 2 black, 1 white and 1 green. In how many ways can 3 beads be chosen from the box and arranged in a row?
(NB. To infer that the beads can only be distinguished by colour.)

Solution

Cases

No. of ways

XXX (3 same colour)

XXY (2 same colour, 1 distinct)

XYZ (3 distinct colours)

Note: The number of ways is 1 for choosing 3 red.

In what case would it be 7P_3 ?

$\therefore$ Total number of ways = $1 + 18 + 24 = 43$

After every lecture, *reflect on what you have learnt.*

Use this space to jot down *salient points*, *important skills*, *key concepts* and *questions that you would like to clarify*. Use them to help you craft *your own summary*.

Summary

Basic Counting Principles

1 Addition Principle

When two operations are **mutually exclusive**, where one can happen in r ways and the other in s ways, then one operation or the other can be done in $r + s$ ways.

2 Multiplication Principle

If a first operation can be performed in r ways, and when it has been performed in any one of these ways, a second operation can then be performed in s ways, then the number of ways of performing the two operations **successively** is $r \times s$.

Example

8, 10(ii), 17, 23

10, 11, 12

Useful Techniques

1 Complement Method

Use it when direct computation is tedious.

Note: Example 5(a) and example 5(c) are **not** complementary events.

5(d), 7(b), 16(i)

2 Two or more objects are adjacent

5(c), 6(a)

3 Separation Method

6(c)

4 Choosing some or all objects (all distinct)

19, 20

5 Choosing some or all objects (not all distinct)

21

6 Choose first, then arrange

22, 23

Computational Expressions

${}^n P_r = {}^n C_r \times r!$ = number of permutations of r objects out of n distinct objects

$n! = {}^n P_n$ = number of permutations of all n distinct objects

$\frac{n!}{p!q!}$ = number of permutations of all n objects, of which p are identical of the first kind, q are identical of the second kind

$1 \times (n-1)!$ = number of permutations of n distinct objects in a circle

${}^n C_r$ = number of combinations of r objects out of n distinct objects

2^n = number of selections of none or any number of objects out of n distinct objects

${}^n C_a \times {}^{n-a} C_b$ = number of selections of a items for the first group, followed by selecting b items from the remaining items for the second group.

$\frac{{}^n C_a \times {}^{n-a} C_a}{2!}$ = number of selections of a items for one group and a items for another group, with both groups non-distinguishable.